

Managing a Three-Year Water and Sanitation Project for Climate Resilience

Climate change has dried up water points and collapsed sanitation systems in many communities, threatening health and livelihoods. As project manager for a UNICEF-funded water and sanitation initiative in our community, I will deliver a sustainable three-year project that builds resilience against extreme weather. The project will restore reliable water access, improve sanitation, and strengthen community capacity for long-term management. Success will be measured not just by infrastructure built, but by systems that continue functioning beyond the funding period.

The project begins with a six-month planning and assessment phase in Year One. I will conduct a baseline survey to map current water sources, sanitation coverage, and climate risks such as drought and flooding. Community participation is essential—I will facilitate inclusive meetings with women, youth, and vulnerable groups to identify needs and priorities. Using UNICEF's sustainability framework, we will assess technical, financial, institutional, and social factors. This includes hydrogeological studies to site new boreholes away from flood-prone areas and water quality testing to ensure safety. Partnerships will be formed with local government, traditional leaders, and technical experts to align with national WASH policies.

Implementation ramps up in the second half of Year One and continues through Year Two. We will construct climate-resilient water points—deep boreholes with hand pumps, protected springs, and rainwater harvesting systems equipped with first-flush diverters. Sanitation facilities will include ventilated improved pit latrines and handwashing stations designed to withstand heavy rains. All infrastructure will incorporate adaptation measures like elevated platforms in flood zones and drought-resistant designs. Community members will be trained in construction and actively involved to build ownership from the start.

Hygiene promotion runs parallel to hardware installation. Using participatory methods, we will deliver behavior change campaigns on handwashing, safe water storage, and open defecation-free practices. School WASH programs will target children, who often act as change agents in households. I will recruit and train local health promoters to sustain these messages beyond the project.

Year Three shifts focus to monitoring, evaluation, and handover for sustainability. A robust monitoring system with monthly sustainability checks will track functionality rates, water quality, and usage. Community Water and Sanitation Committees, with at least 50% women representation, will be established and trained in operation and maintenance, financial management, and conflict resolution. We will set up a maintenance fund through affordable user tariffs and link committees to local spare parts suppliers. Regular climate risk assessments will inform adaptive management—if drought patterns intensify, we can adjust by expanding rainwater systems or groundwater recharge.

Throughout the three years, I will apply results-based management: quarterly progress reports to UNICEF, adaptive planning to respond to emerging challenges, and value-for-money procurement to maximize impact. Gender and equity will be mainstreamed, ensuring facilities are accessible to people with disabilities and that women participate meaningfully in decision-making.

By project close, the community will have functional, climate-resilient WASH infrastructure serving at least 80% of households, trained local committees ready to manage services, and

established monitoring systems. This approach ensures the project does not end when funding stops—it builds lasting resilience against future climate shocks.

In conclusion, effective project management for this UNICEF initiative requires strong community ownership, integration of climate adaptation from the design stage, and a clear focus on sustainability from day one. By following a structured pathway of assessment, implementation, and handover with continuous monitoring, we will deliver water and sanitation systems that endure. This not only cushions the immediate impacts of climate change but equips the community to thrive despite it.